AI-Powered Network Intrusion Detection System with Explainable Machine Learning and Adaptive Threat Scoring

Project Domain: Artificial Intelligence and Cybersecurity

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Abstract

This project explores the application of supervised machine learning techniques for network intrusion detection using selected traffic records from the CICIoT2023 dataset.

A Random Forest classifier was developed to distinguish benign network traffic from DDoS UDP flood traffic. The project includes data inspection, duplicate removal, missing-value handling, preprocessing, model training, performance evaluation, feature importance analysis, feature-ablation experiments, and an initial threat-scoring mechanism.

A separate TCP flood dataset was also examined using the trained models for exploratory prediction analysis. However, this dataset did not contain verified row-level labels, so its predictions cannot be treated as confirmed detection results.

Although the Random Forest achieved 100% metrics on the current random test split, further validation is necessary to determine whether the model generalizes to unseen network environments.

1. Introduction

Network security is an important part of modern computing infrastructure. Networks are continuously exposed to malicious activities, including Distributed Denial of Service (DDoS) attacks.

Network Intrusion Detection Systems (NIDS) monitor network traffic and attempt to identify suspicious or malicious activities.

Machine learning can help identify patterns in network traffic and support automated intrusion detection.

This project investigates the use of machine learning to classify benign and malicious network traffic using selected CICIoT2023 dataset records.

2. Problem Statement

Traditional network monitoring approaches may struggle to identify complex traffic patterns efficiently.

The objective of this project is to investigate whether machine learning can distinguish benign network traffic from DDoS UDP flood traffic and provide an initial threat-scoring mechanism.

3. Research Objectives

- Develop a machine-learning-based network intrusion detection prototype.
- Preprocess and clean network traffic data.
- Train a Random Forest classification model.
- Evaluate the model using standard classification metrics.
- Investigate feature importance and feature dependencies.
- Explore predictions on a separate TCP flood dataset.
- Develop an initial model-score-based threat indicator.

4. Dataset Description

The project uses selected CSV files from the CICIoT2023 dataset.

The main dataset contains:

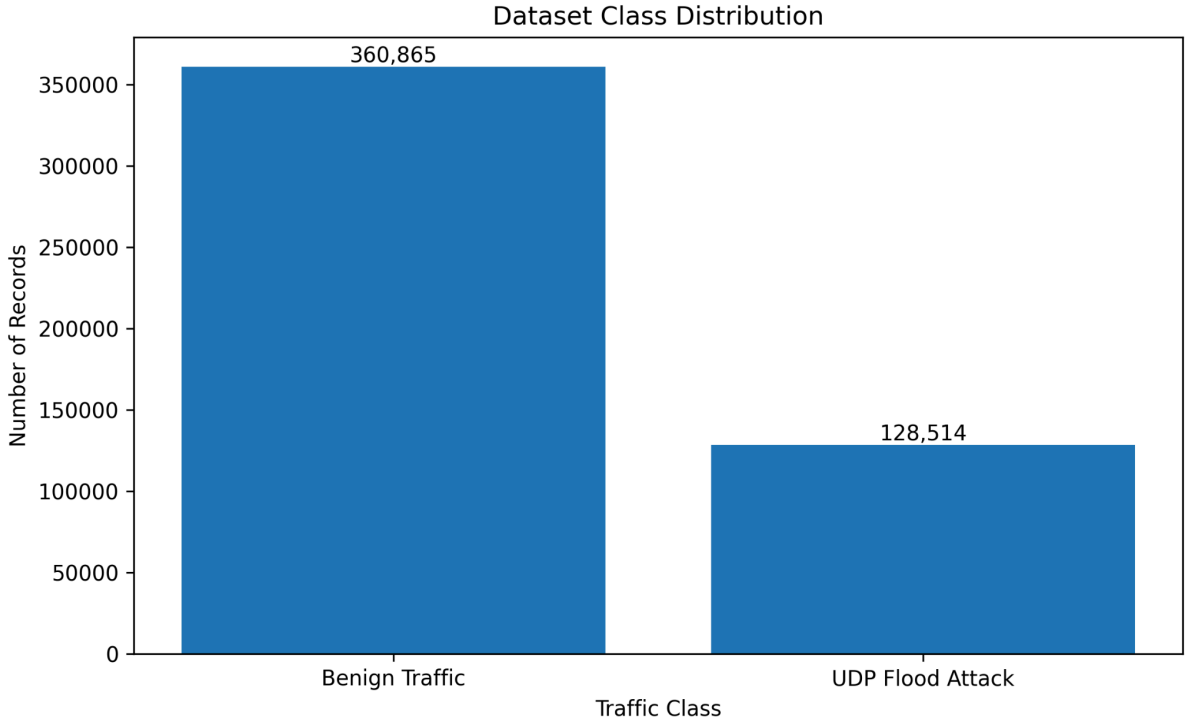
Dataset	Original Rows	Rows After Duplicate Removal
Benign traffic	362,361	360,865
DDoS UDP flood	266,605	128,514
Combined dataset	—	489,379

The dataset contains 39 input features.

The target labels were assigned using the source-file categories:

- Benign traffic = 0
- UDP flood traffic = 1

A separate TCP flood file containing 265,776 records was used for exploratory prediction analysis.



5. Data Preprocessing

The following preprocessing operations were performed:

1. Loaded the selected CSV files.
2. Inspected dataset dimensions and columns.
3. Removed duplicate records.
4. Replaced infinite values with missing values.
5. Assigned binary target labels.
6. Split the dataset into training and testing subsets.
7. Applied median imputation.
8. Verified that the processed datasets contained no missing or infinite values.

An 80/20 stratified train-test split was used.

The imputer was fitted only on the training data to avoid using test-set information during preprocessing.

6. Machine Learning Model

A Random Forest classifier was selected for the classification task.

The model configuration was:

- Algorithm: Random Forest Classifier
- Number of estimators: 100
- Random state: 42
- Parallel processing: Enabled

The model was trained to distinguish benign traffic from DDoS UDP flood traffic.

7. Experimental Results

The model achieved the following results on the random test split:

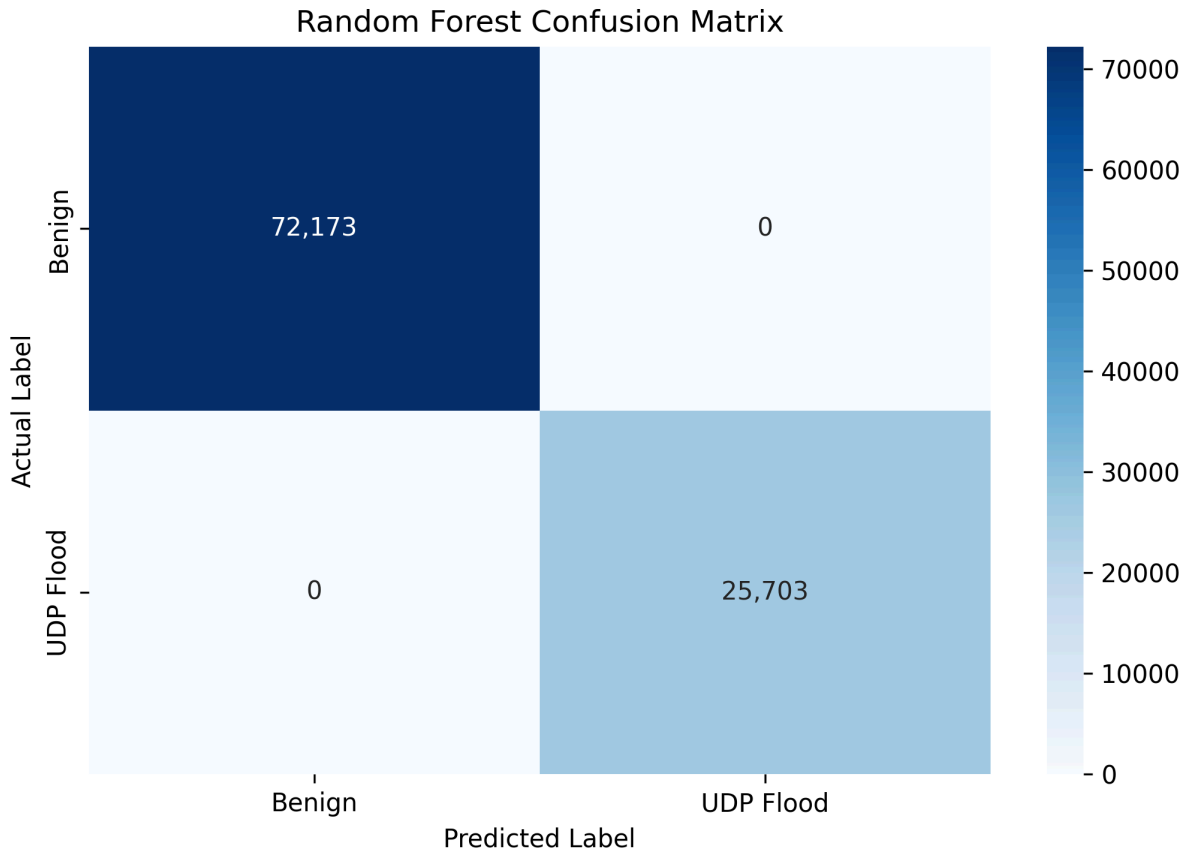
Evaluation Metric	Result
Accuracy	100%
Precision	100%
Recall	100%
F1-score	100%
Balanced Accuracy	100%

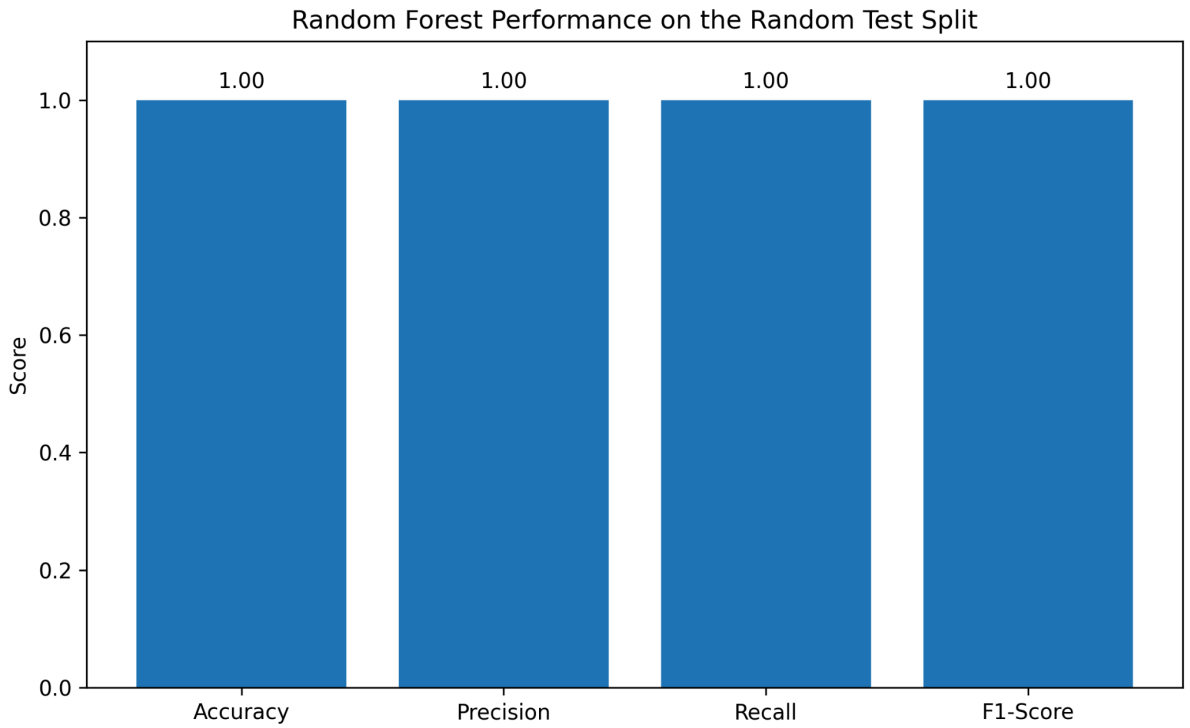
Confusion Matrix

```
[[72173     0]
 [    0 25703]]
```

These results indicate perfect classification on this particular test split.

However, they do not establish 100% real-world detection accuracy. Additional experiments using unseen capture sessions and independently collected network traffic are necessary.





8. Feature Importance Analysis

Feature importance was examined to investigate which attributes influenced the Random Forest model.

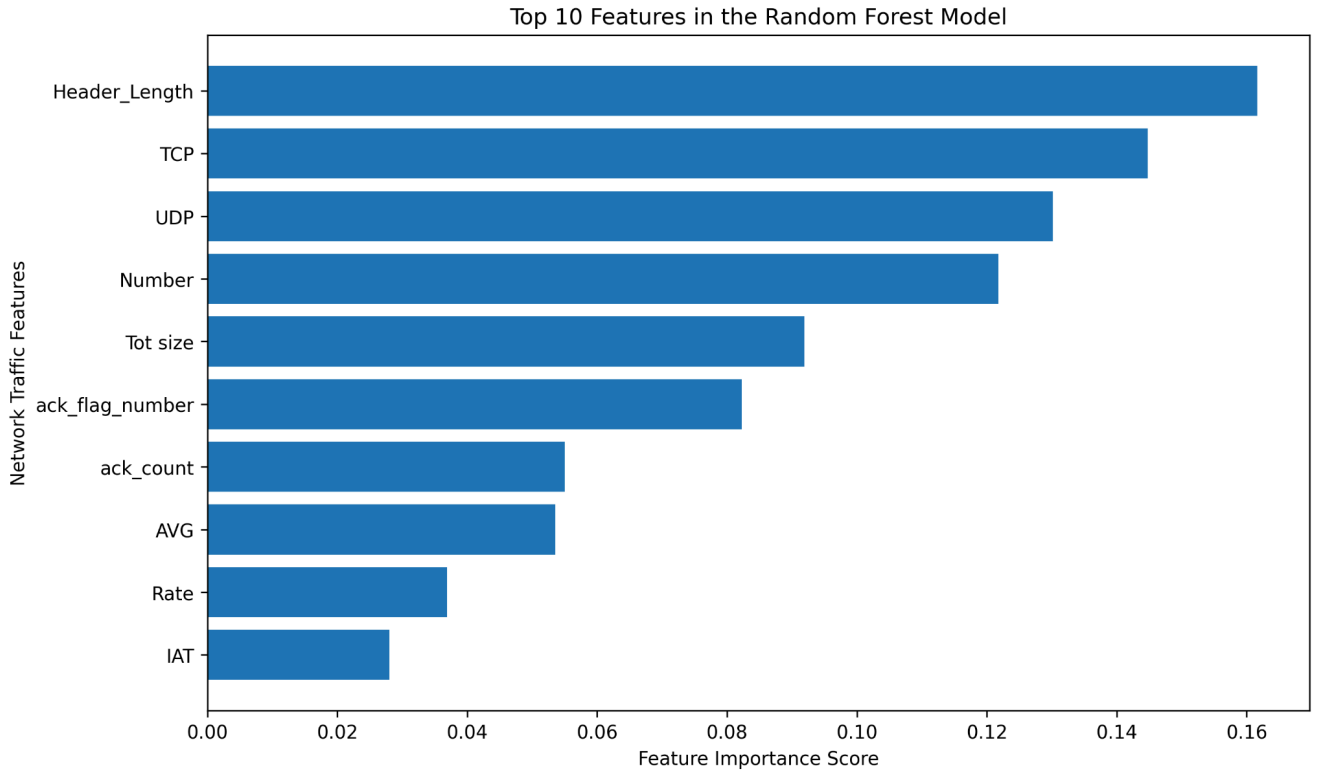
Some of the higher-ranked features included:

- Header_Length
- TCP
- UDP
- Number
- Tot size
- ack_flag_number
- ack_count
- AVG
- Rate
- IAT

The Number feature showed a particularly strong association with the source-file categories.

Approximately 99.95% of benign records had Number = 10, while approximately 99.92% of UDP flood records had Number = 100.

This raises the possibility of dataset-specific shortcuts influencing the classification process.



9. Feature-Ablation Experiments

Additional experiments were performed by removing selected features.

The experiments included:

- Removing Number.
- Removing protocol-related features.
- Removing both Number and protocol-related features.

The Random Forest continued to achieve perfect metrics on the same random test split.

These results require further investigation because other correlated features may still contain dataset-specific information.

10. Exploratory TCP Flood Analysis

A separate TCP flood CSV file was passed through the trained models.

The main model predicted approximately 99.9759% of its records as attacks.

However, the TCP dataset did not contain verified row-level labels in this workflow.

Therefore, these numbers represent model predictions, not confirmed classification accuracy.

Differences between models with different feature sets also indicate that further generalization testing is necessary.

11. Initial Threat Scoring

An initial threat score was calculated using the Random Forest attack-class output score:

Threat Score = Attack-class model score × 100

Illustrative categories were:

- Low: Below 30
- Medium: 30 to below 70
- High: 70 or above

These thresholds are preliminary and have not been calibrated against real-world security incidents.

The score should therefore be considered an experimental prototype rather than a production-ready security risk measurement.

12. Limitations

The project has several limitations:

1. The main classification experiment covers selected traffic categories.
2. Labels were assigned from source-file categories.
3. Random splitting may not fully test generalization to unseen network environments.
4. Perfect test metrics may be influenced by dataset-specific patterns.
5. The Number feature showed a strong association with source-file identity.
6. The separate TCP flood dataset lacked verified ground-truth labels.
7. Threat-scoring thresholds have not been validated in real network environments.

13. Future Work

Future improvements may include:

- Testing on independently collected network traffic.
- Using time-based or device-based train-test splitting.
- Adding more network attack categories.
- Applying SHAP for explainable machine learning.

- Calibrating model output scores.
- Improving the threat-scoring mechanism.
- Developing a real-time network traffic monitoring prototype.

14. Conclusion

This project developed an experimental machine-learning pipeline for network intrusion detection using selected CICIoT2023 traffic records.

The workflow included dataset cleaning, preprocessing, Random Forest model training, classification evaluation, feature importance analysis, feature-ablation experiments, and an initial threat-scoring mechanism.

Although the model achieved perfect metrics on the current random test split, further validation is necessary before making claims about real-world detection performance.

The investigation of dataset-specific feature patterns highlighted the importance of careful experimental design and generalization testing in cybersecurity machine learning research.

15. References

1. Canadian Institute for Cybersecurity, University of New Brunswick. CICIoT2023 Dataset. https://www.unb.ca/cic/datasets/iotdataset-2023.html
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